

PATIENT HISTORY:

CHIEF COMPLAINT/PRE-OP/POST-OP DIAGNOSIS: Thyroid nodule
 PROCEDURE: Not answered
 SPECIFIC CLINICAL QUESTION: Not answered
 OUTSIDE TISSUE DIAGNOSIS: Not answered
 PRIOR MALIGNANCY: Not answered
 CHEMOTHERAPY: Not answered
 ORGAN TRANSPLANT: Not answered
 IMMUNOSUPPRESSION: Not answered
 OTHER DISEASES: Not answered

UUID: BF24324C-EE6E-40E8-BF89-E42D0D02DD6A
 TCGA-BJ-A190-01A-PR

Redacted



ADDENDA:
Addendum

ICD-0-3

carcinoma, papillary, follicular
 variant 8340/3

MOLECULAR ANATOMIC PATHOLOGY TESTING:

Block 1B:

- A. *HRAS* codon 61 mutation IDENTIFIED.
- B. Mutations in *BRAF*, *NRAS*61/*KRAS*12/13 NOT identified.
- C. FISH test results are negative for *RET/PTC* rearrangement.

Site: thyroid, NOS C73.9

Block 1J:

- A. Mutations in *BRAF*, *NRAS*61, *HRAS*61, *KRAS*12/13 NOT identified.
- B. FISH test results are negative for *RET/PTC* rearrangement.

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 6/10/12

NOTE:

DNA was extracted in the amount sufficient for testing.

BACKGROUND:

Mutations in either *BRAF* or *RAS* genes or *RET/PTC* rearrangements are found in more than 70% of papillary thyroid carcinomas (1). *BRAF* V600E (T1799A) mutation has been associated with more aggressive behavior of papillary carcinoma (2, 3). The association between *BRAF* V600E mutation and features of tumor aggressiveness have also been observed in papillary microcarcinomas (4). Mutations in the *RAS* genes or *PAX8/PPARγ* rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (5). Regarding the specificity of these mutations for cancer, *BRAF* V600E mutation and *RET/PTC* and *PAX8/PPARγ* rearrangements are overall specific for malignancy in the thyroid, although they have been reported with a very low frequency in benign thyroid lesions (6). *RAS* mutations occur in malignant and benign thyroid tumors, being found in ~40-60% of follicular and anaplastic carcinomas, 30-40% of follicular adenomas and 10-15% of papillary carcinomas (6).

1. Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. *Am J Surg Pathol* 2006; 30:216-222
2. Xing M, et al. *BRAF* mutation predicts a poorer clinical prognosis for papillary thyroid cancer. *J Clin Endocrinol Metab* 2005; 90:6373-6374.
3. Elisei R, et al. *BRAF* V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. *J Clin Endocrinol Metab* 2008; 93:3943-3949.
4. Lee X, Gao M, Ji Y, Yu Y, Feng Y, Li Y, Zhang Y, Cheng W, Zhao W. Analysis of differential *BRAF*(V600E) mutational status in high aggressive papillary thyroid microcarcinoma. *Ann Surg Oncol*. 2009 Feb;18(2):240-5.
5. Nikiforova MN, et al. *RAS* point mutations and *PAX8-PPARγ* rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. *J Clin Endocrinol Metab* 2003; 88: 2318-26.
6. Nikiforov YE. Recent developments in the molecular biology of the thyroid. In: Lloyd RV, ed. *Endocrine Pathology: Differential Diagnosis and Molecular Advances*. Humana Press, Totowa, 2004, 191-208.

MUTATIONAL ANALYSIS:

For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the *rt* platform to amplify *BRAF* codons 599-601, *NRAS* codon 61, *HRAS* codon 61, and *KRAS* codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. If required, the mutation type was confirmed by sequencing of the PCR product on DNA from samples positive for each of these mutations was used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10-20% of alleles with mutation present in the background of normal

FINAL DIAGNOSIS:

THYROID, TOTAL THYROIDECTOMY (34.9 GRAMS) -

- A. PAPILLARY THYROID CARCINOMAS, ENCAPSULATED FOLLICULAR VARIANTS, TWO FOCI, INVOLVING RIGHT LOBE (1.0 CM), AND LEFT LOBE (3.5 CM) (see comment).
- B. NO EXTRACAPSULAR EXTENSION OR ANGIOLYMPHATIC INVASION.
- C. TUMORS CONFINED TO THE THYROID.
- D. ONE LEFT PERITHYROIDAL LYMPH NODE, NO TUMOR PRESENT (0/1).
- E. PATHOLOGIC STAGE: pT2N0.

Criteria	Yes	No
Diagnosis Discrepancy		
Primary Tumor Site Discrepancy		
HIPAA Discrepancy		
Prior Malignancy History		
Dual/Synchronous Primary Noted		
Case Is (circle): QUALIFIED / DISQUALIFIED		
Reviewer Initials	mg	6/10/12

COMMENT:

Immunohistochemical stains were performed on Blocks D, F and J. The results are as follows:

Block B: HBME-1 focal membranous staining
 Galectin 3 - negative.

Block F: HBME-1 - negative
 Galectin 3 - negative.

Block J: HBME-1 - negative
 Galectin 3 - negative.

These immunostaining results and the 'borderline' nuclear features of the two most suspicious nodules (represented in block B, right and block J, left) have made this quite a challenging case requiring additional consultation. Thus, this case was reviewed by [redacted] For each of these nodules, a majority opinion was reached reflected in the diagnosis above.

CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE:	Total Thyroidectomy
TUMOR SITE:	Right Lobe, Left Lobe
TUMOR FOCALITY:	Multifocal
TUMOR SIZE (largest nodule):	Greatest Dimension: 3.5 cm
HISTOLOGIC TYPE**:	Papillary carcinoma, encapsulated follicular variant
TNM-DESCRIPTORS:	m
PRIMARY TUMOR (pT):	pT2
REGIONAL LYMPH NODES (pN):	pN0
	Number of regional lymph nodes examined: 1
	Number of regional lymph nodes involved: 0
EXTRANODAL EXTENSION:	Not identified
DISTANT METASTASIS (pM):	Not applicable
EXTRATHYROIDAL EXTENSION:	Not identified
MARGINS:	Margins uninvolved by carcinoma
LYMPH-VASCULAR INVASION:	Not identified
ADDITIONAL PATHOLOGIC FINDINGS:	None Identified

SPECIAL PROCEDURES:

In Situ Procedure

Interpretation

RESULTS: Two blocks (1B & 1J) were analyzed with the following results:

1B: RET/PTC: The targeted area of the tissue showed 0(0%) of the cells with the rearrangement pattern and 62(100%) of the cells with the normal pattern. The targeted area is considered NOT rearranged for the RET/PTC region.

The RET/PTC1 Probe did not contain the rearrangement pattern.

1J: RET/PTC: The targeted area of the tissue showed 0(0%) of the cells with the rearrangement pattern and 60(100%) of the cells with the normal pattern. The targeted area is considered NOT rearranged for the RET/PTC region.

The RET/PTC1 Probe did not contain the rearrangement pattern.

Results

RET/PTC Methodology Description:

Probe: RET/PTC Probe (10q11.2) and RET/PTC1 Dual-Color Fusion Probe 10q11.2-10q21

Probe Description: The RET probe (labeled SpectrumGreen) is a 207 kb DNA probe located at band 10q11.2.

The H4 and 369 probes are approximately 100kb each (labeled in SpectrumOrange), and are located at band 10q21.

RET/PTC rearrangement is suggested to be present when three RET signals are identified in greater than 8% of the analyzed cells.

RET/PTC1 rearrangement detection is as follows: Normal cells without the RET/PTC1 rearrangement show two pair of green and orange signals separated. Cells with the RET/PTC1 rearrangement show two pair of signals (green and orange) in juxtaposition and one green and one orange signal separated in greater than 8% of analyzed cells within the targeted area.

RET/PTC FISH analysis was manually performed and quantitatively assessed by analysis of a minimum of 60 cells using the PTC (369/H4) SpectrumOrange and the RET SpectrumGreen probes.